

House Price Prediction using Advanced Regression Techniques

Research Report

1. Introduction

Accurately predicting house prices is a problem of significant practical value to buyers, sellers, real estate agents, investors, and policymakers. A reliable predictive model allows buyers to identify fair market value, helps sellers price competitively, and gives investors a data-driven basis for portfolio decisions. This report documents the research, methodology, and findings behind a machine learning system built to predict residential sale prices using advanced regression techniques, applied to the Ames Housing dataset.

The project moves beyond simple linear regression to explore regularized linear models, ensemble tree-based methods, hyperparameter optimization, and model explainability — techniques that reflect current practice in applied data science rather than introductory coursework.

2. Definitions

2.1 Data Analytics

Data Analytics refers to the process of examining datasets to draw conclusions about the information they contain, typically using statistical and visualization techniques to identify patterns, trends, and relationships in historical data.

2.2 Data Science

Data Science is a broader, interdisciplinary field combining statistics, programming, and domain expertise to extract insights and build predictive systems from structured and unstructured data. Where analytics tends to describe what happened, data science aims to also predict what will happen and prescribe what to do about it.

2.3 Artificial Intelligence

Artificial Intelligence (AI) is the broader discipline concerned with building systems capable of performing tasks that normally require human intelligence. Machine learning, the family of techniques used in this project, is a subset of AI in which systems learn patterns from data rather than following explicitly programmed rules.

2.4 Advanced Regression

Advanced regression refers to techniques that extend ordinary linear regression to handle non-linearity, multicollinearity, and high-dimensional feature spaces — including regularized methods (Ridge, Lasso), polynomial feature expansion, and ensemble tree-based regressors (Random Forest, XGBoost).

3. Role of Data Science in Real Estate

Real estate pricing is influenced by a large number of interacting factors: physical attributes of a property (size, quality, age), location-based factors (neighborhood, proximity to amenities), and macroeconomic conditions (interest rates, local market demand). Traditional appraisal relies heavily on manual comparison to similar recently-sold properties. Data science offers a systematic alternative: a model can simultaneously weigh dozens of

features and learn relationships that would be difficult for a human appraiser to quantify consistently, offering a scalable and reproducible approach to price estimation.

4. Dataset

This project uses the Ames Housing dataset, compiled by Dean De Cock as a modernized alternative to the frequently used Boston Housing dataset. The Boston Housing dataset was deliberately avoided due to well-documented ethical concerns, including a feature that encoded a proxy for racial demographics — its removal from mainstream libraries such as scikit-learn reflects a broader shift in the field toward more responsible dataset selection.

The Ames dataset contains 1,460 training observations and 1,459 test observations, each described by 79 explanatory variables covering nearly every aspect of residential homes sold in Ames, Iowa between 2006 and 2010, along with the target variable, SalePrice.

5. Methodology

5.1 Exploratory Data Analysis

Initial analysis revealed that SalePrice is right-skewed (skewness = 1.88), motivating a log1p transformation of the target prior to model training. Correlation analysis identified OverallQual ($r = 0.79$) and GrLivArea ($r = 0.71$) as the strongest individual predictors of price.

5.2 Outlier Treatment

Rather than applying blanket statistical outlier removal across all numeric columns — which risked discarding a meaningful portion of an already modest 1,460-row dataset — outliers were removed selectively. Two properties with GrLivArea exceeding 4,000 sq. ft. but unexpectedly low sale prices were identified as genuine anomalies that contradicted the expected size-to-price relationship, and were removed prior to training.

5.3 Missing Value Imputation

Missing value analysis revealed that the majority of missingness in this dataset is structural rather than incidental: features such as PoolQC, FireplaceQu, and the Garage/Basement column groups record 'NA' to indicate the absence of that feature (e.g., no pool), not a data collection error. These were imputed with an explicit 'None' category. Genuinely missing values, such as LotFrontage, were imputed using the median value grouped by neighborhood, reflecting the fact that lot frontage varies systematically by location.

5.4 Feature Engineering

Several domain-informed features were engineered to strengthen predictive signal, including TotalSF (combined basement, first-floor, and second-floor area), TotalBath (a weighted combination of full and half bathrooms), HouseAge, and YearsSinceRemodel. TotalSF achieved a correlation of 0.83 with SalePrice, surpassing every individual predictor in the raw dataset, confirming the value of combining related features.

5.5 Encoding and Transformation

Categorical variables were one-hot encoded, expanding the feature space to 269 columns. Numeric features exhibiting high skewness ($|\text{skew}| > 0.75$) were log-transformed to better satisfy the assumptions of linear model residuals.

6. Advanced Regression Techniques Applied

6.1 Ridge Regression

Ridge regression applies an L2 penalty to model coefficients, shrinking them toward zero without eliminating them entirely. This is particularly effective when predictors are correlated with one another, as is the case here — several engineered and raw features (e.g., GarageCars and GarageArea) exhibit meaningful multicollinearity.

6.2 Lasso Regression

Lasso regression applies an L1 penalty, which can shrink some coefficients to exactly zero, effectively performing feature selection. In this project, Lasso achieved the second-best individual model performance (RMSE 0.1194), narrowly outperforming Ridge.

6.3 Random Forest

Random Forest builds an ensemble of decision trees trained on bootstrapped samples and random feature subsets, averaging their predictions to reduce variance. In its default configuration it underperformed relative to the linear models (RMSE 0.1474), likely due to the high-dimensional, sparse feature space created by one-hot encoding.

6.4 XGBoost

XGBoost is a gradient-boosted tree ensemble that builds trees sequentially, each correcting the errors of the previous ensemble. In its default configuration it underperformed the linear models, but after hyperparameter tuning it became the strongest individual model (RMSE 0.1190).

6.5 Stacked Ensemble

A stacking ensemble was constructed using Ridge, Lasso, and tuned XGBoost as base learners, with a Ridge meta-model trained (via 5-fold cross-validation) to learn the optimal weighted combination of their predictions. This ensemble achieved the best overall performance (RMSE 0.1155), confirming that linear and tree-based approaches capture complementary aspects of the underlying relationships.

7. Feature Selection and Hyperparameter Tuning

Hyperparameter optimization for XGBoost was performed using Optuna, a Bayesian-informed optimization framework that intelligently searches the hyperparameter space rather than exhaustively testing a fixed grid, as GridSearchCV would. Across 50 trials, validation RMSE improved from 0.1276 (default parameters) to 0.1190. Notably, the optimal configuration favored a shallow tree depth (`max_depth = 2`), suggesting that the relationships between features and SalePrice in this dataset are largely simple and near-linear — a finding consistent with the strong performance of the regularized linear models.

8. Model Evaluation and Results

All models were evaluated on a held-out validation set (20% of training data) using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), both computed on the log-transformed target, alongside R^2 as a measure of explained variance. Results are summarized below.

Model	RMSE	MAE	R^2
Linear Regression	0.1344	0.0881	0.8928
Random Forest	0.1474	0.0984	0.8712
XGBoost (Untuned)	0.1276	0.0871	0.9035
Ridge	0.1198	0.0821	0.9149

Lasso	0.1194	0.0820	0.9154
XGBoost (Tuned)	0.1190	0.0822	0.9160
Stacked Ensemble	0.1155	0.0784	0.9209

The stacked ensemble outperformed every individual model, confirming the value of combining diverse learning approaches over relying on any single algorithm.

9. Model Explainability (SHAP)

SHAP (SHapley Additive exPlanations) values were computed for the tuned XGBoost model to address the interpretability limitations typically associated with tree-based ensembles. The SHAP summary plot confirmed TotalSF and OverallQual as the dominant drivers of predicted price, consistent with earlier correlation analysis — validating that the model's learned behavior aligns with domain intuition rather than relying on spurious patterns. Individual-level SHAP force plots further demonstrated the ability to decompose a single prediction into per-feature contributions, providing transparency that a black-box prediction alone would not offer.

10. Literature Review

- De Cock, D. (2011). Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project. The original dataset publication, positioning Ames Housing as a modernized, pedagogically richer alternative to Boston Housing.
- A comparative study evaluating Linear Regression, MLP, Random Forest, SVR, and XGBoost on this dataset found XGBoost achieved the highest R^2 (0.93), and performed strongly even prior to hyperparameter tuning — consistent with this project's own findings after tuning.
- A separate study applying LASSO, Elastic Net, Gradient Boosting, XGBoost, LightGBM, and a stacked model under five-fold cross-validation reinforced the value of ensemble stacking on this exact dataset.
- Multiple practitioner analyses of this dataset note that many categorical fields use 'NA' as a valid category (e.g., no fireplace) rather than a genuinely missing value — an insight directly incorporated into this project's imputation strategy.
- A hyperparameter-tuned Ridge/Lasso approach on this dataset reported an R^2 of approximately 0.928 after addressing missing data, skewness, and multicollinearity — providing an external benchmark broadly consistent with this project's own linear model results.

11. Domain-Specific Relevance

Housing markets are inherently complex systems shaped by physical property characteristics, neighborhood-level effects, and broader economic conditions. This complexity is reflected in the dataset's 79 original features spanning structural, locational, and condition-based attributes. The relatively strong performance of near-linear models (Ridge, Lasso) alongside the finding that optimal tree depth was shallow suggests that, within a single, geographically-contained market such as Ames over a fixed time window, price relationships are largely additive rather than deeply non-linear — though this may not generalize to more heterogeneous markets or time periods spanning significant economic shifts.

12. Ethical Considerations

The choice of dataset was informed by ethical considerations from the outset: the Boston Housing dataset was avoided due to a documented feature encoding racial demographic data as a proxy variable, a well-known issue that led to its removal from scikit-learn. More broadly, any model trained on historical housing data risks

encoding existing societal biases — for instance, neighborhood-based features can act as indirect proxies for socioeconomic or demographic factors. Deploying such models in real-world lending or appraisal contexts would require careful auditing for disparate impact, a consideration acknowledged here even though it falls outside the immediate scope of this academic project.

13. Conclusion

This project demonstrated an end-to-end advanced regression pipeline for house price prediction, progressing from exploratory analysis and principled data cleaning through feature engineering, multiple regression paradigms, hyperparameter optimization, and model explainability. A stacked ensemble of Ridge, Lasso, and tuned XGBoost achieved the strongest performance (RMSE 0.1155, R^2 0.9209), outperforming every individual model. The results reinforce a broader lesson in applied machine learning: model performance depends as much on principled data preparation and thoughtful feature engineering as on the choice of algorithm itself.

References

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